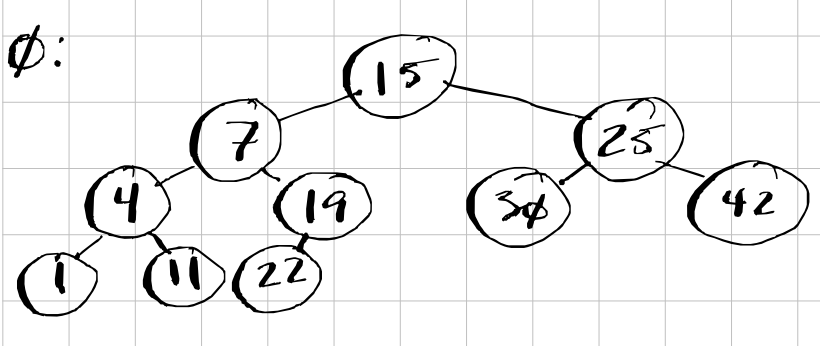
Homework 4

(Tripp) Milton Lamb CS617 Fall 2026

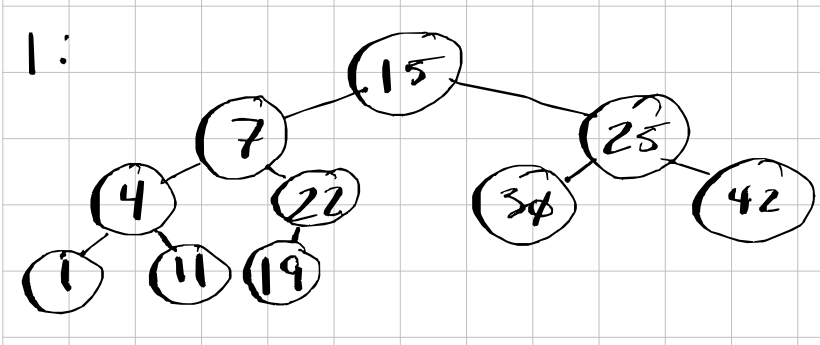
Problem 1

{15, 7, 25, 4, 19, 30, 42, 1, 11, 22}

0:



1:



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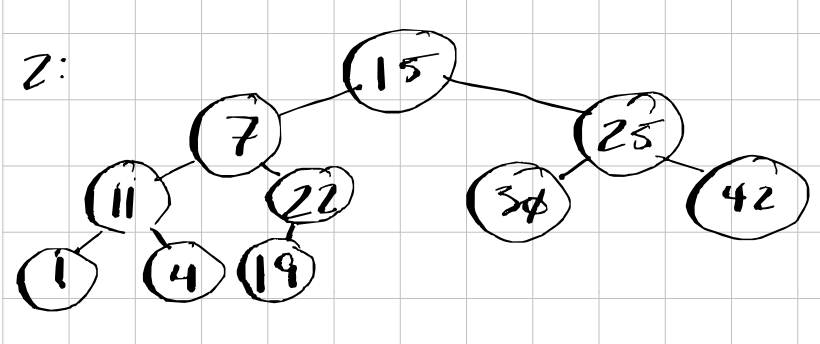

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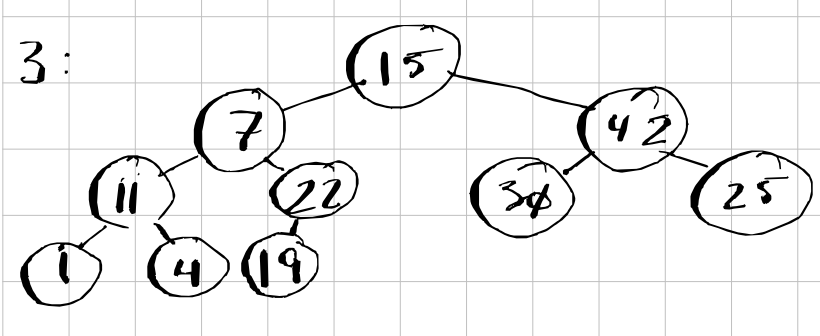

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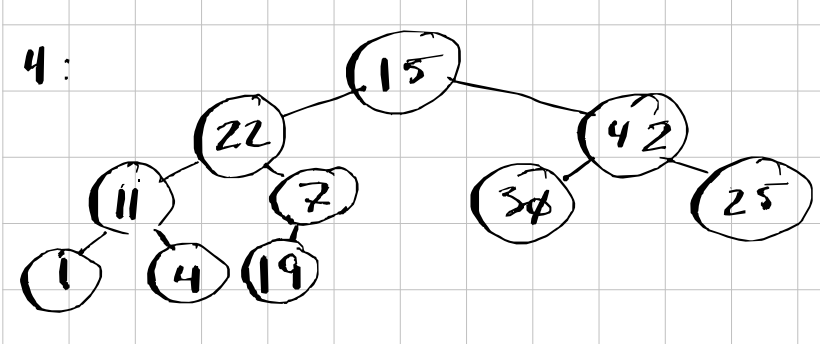

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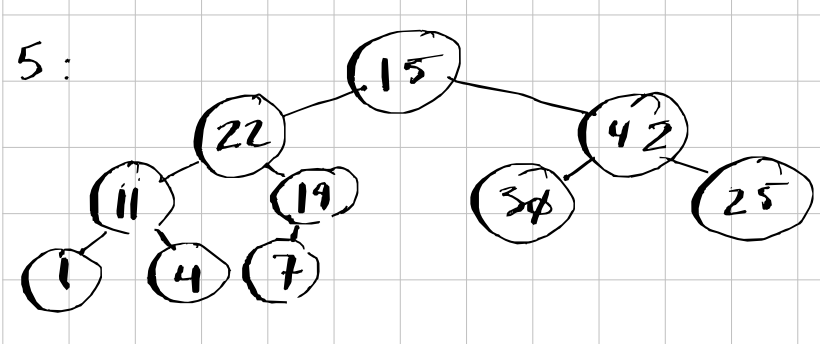

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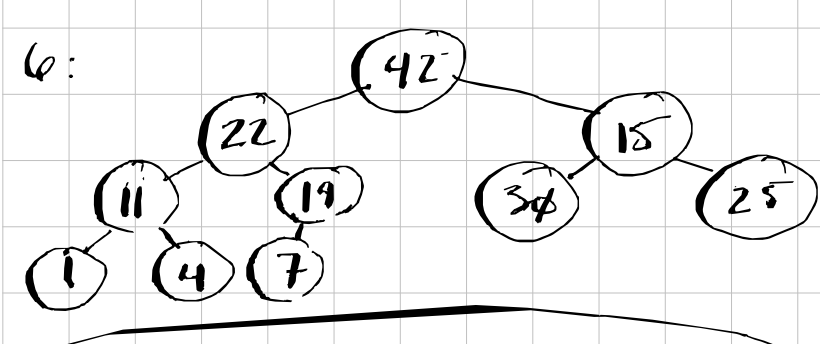
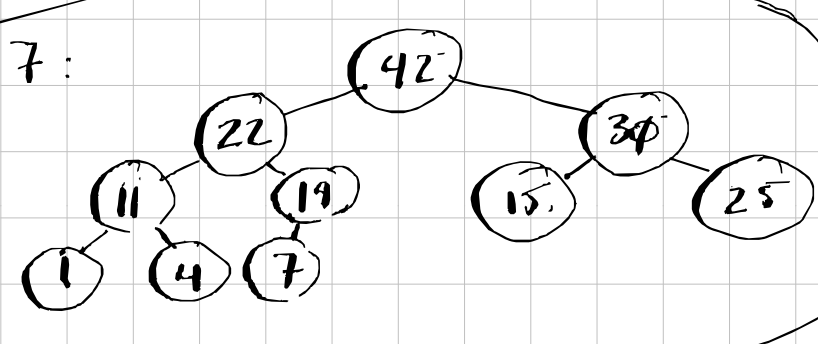


7:



Problem 2

$$A = [2, 5, 3, 0, 2, 3, 0, 3, 1, 4]$$

$$0 \le k \le 5$$

a) $c = [\overset{0}{2}, \overset{1}{1}, \overset{2}{2}, \overset{3}{3}, \overset{4}{1}, \overset{5}{1}]$

b) $c' = [2, 3, 5, 8, 9, 10]$

c) $A' = [\overset{1}{0}, \overset{2}{0}, \overset{3}{1}, \overset{4}{2}, \overset{5}{2}, \overset{6}{3}, \overset{7}{3}, \overset{8}{3}, \overset{9}{4}, \overset{10}{5}]$

$c'' = [0, 2, 3, 5, 8, 9]$

Problem 3

$$m[i,j] = \begin{cases} 0 & \text{if } i = j \\ \min\limits_{i \le k < j} \{ m[i,k] + m[k+1,j] + P_{i-1} P_k P_j \} & i < j \end{cases}$$

$$\underset{P_0}{4} \quad \underset{P_1}{3} \quad \underset{P_2}{7} \quad \underset{P_3}{2} \quad \underset{P_4}{9}$$

$$A_1 \times A_2 \times A_3 \times A_4$$

	1	2	3	4
1	0	m[1,2] = 84 s(1,2) = 1	m[1,3] = 66 s(1,3) = 1	m[1,4] = 138 s(1,4) = 3
2		0	m[2,3] = 42 s(2,3) = 2	m[2,4] = 96 s(2,4) = 3
3			0	m[3,4] = 126 s(3,4) = 3
4				0

$m[1,2] = 4 \times 3 \times 7 = 84$

$m[2,3] = 3 \times 7 \times 2 = 42$

$m[3,4] = 7 \times 2 \times 9 = 126$

$m[1,3](k=1) = m[1,1] + m[2,3] + 4 \times 3 \times 2 = 0 + 42 + 24 = 66$

$m[1,3](k=2) = m[1,2] + m[3,3] + 4 \times 7 \times 2 = 84 + 0 + 56 = 140$

$m[2,4](k=2) = m[2,2] + m[3,4] + P_1 P_2 P_4 = 0 + 126 + 3 \times 7 \times 9 = 315$

$m[2,4](k=3) = m[2,3] + m[4,4] + P_1 P_3 P_4 = 42 + 0 + 3 \times 2 \times 9 = 96$

$m[1,4](k=1) = m[1,1] + m[2,4] + P_0 P_1 P_4 = 0 + 96 + 4 \times 3 \times 9 = 204$

$m[1,4](k=2) = m[1,2] + m[3,4] + P_0 P_2 P_4 = 84 + 126 + 4 \times 7 \times 9 = 462$

$m[1,4](k=3) = m[1,3] + m[4,4] + P_0 P_3 P_4 = 66 + 0 \quad 4 \times 2 \times 9 = 138$

a)

$m[i,i] = 0$

$m[1,2] = 84$

$m[1,3] = 66$

$m[1,4] = 138$

$m[2,3] = 42$

$m[2,4] = 96$

$m[3,4] = 126$

b)

$s[i,i] = 0$

$s[1,2] = 1$

$s[1,3] = 1$

$s[1,4] = 3$

$s[2,3] = 2$

$s[2,4] = 3$

$s[3,4] = 3$

c)

optimal cost $= m[1,4] = 138$

$(A_1 (A_2 A_3)) A_4$